ASSIGNMENT (EMERGING TOPICS PATH)

Topic: Can generative AI create better visual marketing content than humans?

This assignment will help you understand how AI-generated marketing imagery is perceived compared to human-made images based on a small-scale real-world application.

Background

Artificial intelligence, defined as 'the use of computational machinery to emulate capabilities inherent in humans' (Huang & Rust, 2021b), continues to play a significant role in marketing decisions. In the early stages of AI development, marketers used AI primarily for predicting consumer behaviour—often referred to as 'prediction machines' or 'AI for prediction'—to inform their marketing strategies, such as online recommender systems, programmatic advertising, automated email campaigns, (re)targeting, and social media and sentiment analysis. With the advent of transformers and diffusion models (new machine learning techniques), marketers have shifted focus to generating new content, such as text, images, or videos, through applications like ChatGPT and other large language models (LLMs) that augment, imitate, and even replace human capabilities. This shift is known as 'AI for generation' or 'generative AI.' Within the realm of generative AI, content creation has seen exponential growth, particularly in areas like advertising, new product development, and strategic branding decisions.

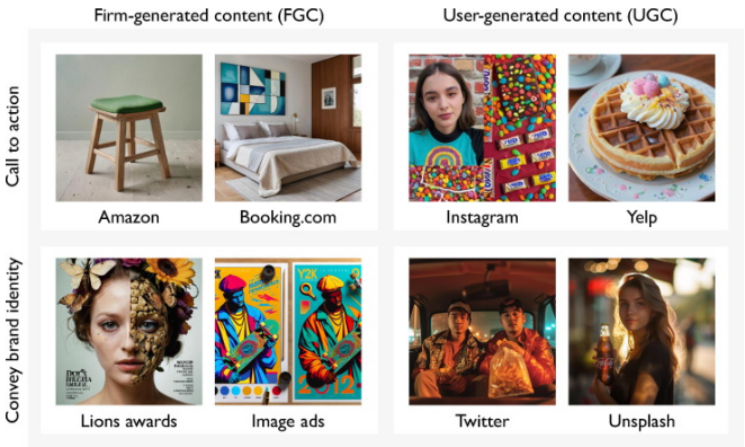
Overview

In this assignment, you will (i) create synthetic marketing images using real-world human-made images as input, and (ii) evaluate whether AI-generated imagery can surpass human-made images in terms of quality, realism, and aesthetics, and (iii) reflect on points (i) and (ii), alongside your session on advertising effectiveness to discuss the advantages and risks of using generative AI for content creation. Specifically, reflect on whether gen AI will take the human out of the equation when it comes to content creation for advertising, new product development, and branding.

Instructions

1. Image collection

Download approximately 30-50 images from Instagram across various brands, e.g., Nike, Airnbn, Coca-Cola, Starbucks etc. These images can be either firm-generated or user-generated and may vary in content, such as 'call to action' images or brand identity visuals. You may want to use the following example as a guide:



Source: Hartmann et al. (2024), "The Power of Generative Marketing: Can Generative AI create superhuman visual marketing content, " International Journal of Research in Marketing.

2. Generate synthetic images

Use free 'text-to-image' gen AI tools to create synthetic images based on the real world data downloaded from Instagram. Suggested tools include:

- *DALL-E 3:* Available for free through Bing's Image Creator and integrated into ChatGPT's interface for some users. Advanced features may require a ChatGPT Plus subscription, but you do not need a subscription for this assignment.
- *Stable Diffusion XL Turbo:* Part of the open-source Stable Diffusion family, generally free to use across various platforms.
- *Realistic Vision:* A model built on Stable Diffusion, available for free on some open-source AI platforms.

3. Conduct a perception survey

Using a short survey, ask your classmates, friends, or family members how they perceive AI-generated images compared to human-made images. Without disclosing whether an image is AI-generated or human-made, assess perceptions on the following variables: **Quality**, **Realism**, and **Aesthetics**.

Use a 7-point Likert scale, from 1 (Strongly Disagree) to 7 (Strongly Agree). Below are sample items for each variable:

Quality

"The image is high quality."

1: Strongly disagree, 2: disagree, 3: somewhat disagree, 4: neutral, 5: somewhat agree, 6: agree, 7: strongly agree.

Realism
"The image looks realistic."
Scale: 1 (Strongly Disagree) to 7 (Strongly Agree)

Aesthetics
"The aesthetic of this image is appealing."
Scale: 1 (Strongly Disagree) to 7 (Strongly Agree)
"The image is visually attractive."
Scale: 1 (Strongly Disagree) to 7 (Strongly Agree)
"The image shows a strong sense of artistic style."
Scale: 1 (Strongly Disagree) to 7 (Strongly Agree)

4. Summarize findings

Analyze your survey results and summarize them using simple descriptive statistics and graphs.

Deliverables

File submissions

Survey: Submit the survey that you used to collect participant responses.

Data analysis: Submit the code (e.g., Python or R) or an Excel spreadsheet used for descriptive statistics and graphs.

Presentation

- Deliver a short presentation summarizing your findings and addressing the key questions specified above.
- Use clear visuals and layouts to help explain your findings.
- Discuss the pros and cons of using gen AI for content creation and address the question of whether gen AI could replace human involvement in content creation for advertising, new product development, and branding.

Peer feedback

After viewing each group's presentation, engage in a feedback session. You are required to provide constructive and critical insights into each group's work.

Assessment criteria:

Clarity and Structure: Is the presentation well-organized, clear, and easy to follow?
Content Insight: Does the presentation thoroughly explain your approach to image generation, survey design, data analysis, and findings on generative AI's role in marketing?
Critical Reflection: Has the group thoughtfully discussed the benefits and challenges of generative AI in marketing and its potential impact on human roles?
Use of Visuals: Are visuals and data presented effectively to support insights?

Reference

J. Hartmann, Y. Exner, S. Domdey, ((2024), "The power of generative marketing: Can generative AI create superhuman visual marketing content?," International Journal of Research in Marketing, doi: https://doi.org/10.1016/j.ijresmar.2024.09.002

ASSIGNMENT (METHODOLOGICAL PATH)

Topic: Marketing analytics dashboard

Background

This assignment gives you the opportunity to work hands-on with data visualization and analytics, building a practical tool that informs high-impact marketing decisions in an e-commerce setting.

As the marketing manager of an e-commerce platform, you are focused on using data-driven insights to better forecast demand, evaluate promotions, and optimize pricing strategies for your product line. You've reached out to your analytics team for a dashboard that offers a comprehensive, visual report addressing critical questions for the next periods' planning.

Objective

Using the provided dataset [https://demandprediction.github.io/dataset.html], your task is to develop an interactive marketing mix dashboard using R or Python that answers the marketing manager's key questions. This dashboard should include predictions, metrics, and visualizations to guide data-driven decision-making related to demand forecasting, promotion effectiveness, and pricing scenarios.

Instructions

1. Tool selection and setup

You will use **Python** (with libraries such as Streamlit, Dash, Plotly, or Voila) or **R** (using Shiny or Flexdashboard) to develop your dashboard. Please ensure the tool you choose supports interactive charts and can display data outputs in a user-friendly format.

2. Demand forecasting

- Question to Address: What is the expected demand for each SKU in the next few periods?
- Implement regression and/or neural network models to predict demand for each SKU in the next periods.
- Create a visual forecast display for each SKU, including confidence intervals if possible.

3. Feature promotion effectiveness

- Questions to Address: Do feature promotions effectively boost demand, and for which SKUs? How much incremental sales are generated from feature promotions?"
- Use the SCAN*PRO model or an alternative approach to assess the impact of feature promotions on demand.
- Measure the lift in sales attributed to feature promotions, presenting results as incremental sales for each SKU.
- Visualize results that indicate which SKUs show a significant increase in demand due to promotions.

4. Price elasticity and scenario testing

- Questions to Address: Which SKUs are most sensitive to price changes? How would a 10% price increase or decrease impact demand and revenue?
- Using a suitable model covered during the module, calculate price elasticity for each SKU and visualize how price changes impact demand.
- Conduct 'what if' scenarios by simulating demand changes under a 5%, 10%, 20%, and 30% price increase and a 5%, 10%, 20%, and 30% price decrease.

Deliverables

1. **Code submission**

    - Submit your Python or R code, including all necessary files and instructions for setting up and running the dashboard locally.
    - Ensure your code is well-commented and structured.

2. **Dashboard and presentation**

    - Design an intuitive, well-organized dashboard that addresses the marketing manager's key questions.
    - Focus on clear visuals and layouts to facilitate data-driven decisions.
    - Provide a link to the dashboard if it is hosted, or include clear instructions on how to run it locally.
    - Create a short presentation that summarizes your findings and addresses each question posed by the marketing manager.

3. **Peer feedback:** After viewing each group's presentation, engage in a feedback session, you are required to provide constructive and critical insights into each group's work.

Assessment criteria

Please make sure that your presentation delivers on (i) the accuracy of demand forecasts, (ii) insights into promotion effectiveness and interpretation of incremental sales, (iii) the ease of price sensitivity analysis, and (iv) the clarity and usability of the dashboard for decision-making.

ASSIGNMENT (AI-TOOL PATH)

Topic: Integrating AI into Retailing for Health and Wellbeing Products

Overview:

This assignment enables you to apply strategic retail principles in a simulated environment, using Planner 5D to visualize and refine a retail layout. It allows you to consider how each design choice—from layout to product placement—affects the shopping experience and reinforces the health and wellbeing message of a new product line on health and wellbeing.

Specifics:

Imagine you are the store manager of a popular retail chain, and you have just been informed that Unilever is launching a new line of health and wellbeing products tailored to current consumer trends. This product line aims to redefine the health and wellness category by aligning with consumer values based on the following trends[1]:

1. *Lifestyle goals:* Consumers no longer see health as the absence of illness but as a lifestyle goal to pursue.
2. *Benefit-led personalization:* Products are shifting from one-size-fits all approach to benefit-led personalization approach.
3. *Overlapping of categories:* Categories such as vitamin and mineral supplements, beauty, personal care, and foods were separate categories but not anymore.

[1] https://www.unilever.com/news/news-search/2022/introducing-our-fastgrowing-health-wellbeing-business/

4. *Sense of community:* The rise of the conscious consumer and rapid shift to digital leads to a sense of community and lifestyle.

Your task is to leverage an AI tool to create a retail environment that aligns with these trends, focusing on effective product placement, customer engagement, and a layout that fosters a sense of lifestyle and community.

Key aspects of the assignment:

1. **AI-driven retail design:**

   - Use Planner 5D to design and visualize a store layout tailored to the health and wellbeing products (please note that you may want to use a different AI tool as well). You'll experiment with different layouts, product placements, and customer interaction points to create an engaging retail environment.

   - Reference the instructions in the "Planner 5D Guide" below to set up and customize your layout. You may want to experiment with elements like lighting, flooring, and wall colours to enhance the store's ambiance and reflect a health and wellness theme. Use your imagination!

2. **Product placement and category blending:**

   - Position the new product line alongside relevant categories like vitamins and supplements (VMS), beauty, personal care, and foods. Use **Planner 5D**'s shelving and display features to visually represent strategic product placements that encourage exploration and create an intuitive shopping experience.

   - Design your layout to showcase the personalization aspect, catering to customers looking for products that meet their unique health and wellness goals.

3. **Customer interaction and engagement:**

   - Identify customer touchpoints within the store layout to encourage interaction. Consider adding display areas or signage that can provide product information or health tips, fostering a sense of community and lifestyle within the store.

   - Integrate digital or AI-driven elements where possible, like simulated recommendation displays or interactive areas, to enrich the customer experience and create a dynamic shopping environment.

   - Think about how to foster a sense of community and lifestyle within the store, creating a shopping experience that goes beyond just products.

Presentation and Feedback:

Group Presentation: Once your AI-driven retail environment design is complete, you will pitch it to the class. Be prepared to explain the rationale behind your design choices, particularly how they meet the evolving needs of today's health-conscious and digitally engaged consumers.

Peer feedback: After viewing each group's presentation, engage in a feedback session, you are required to provide constructive and critical insights into each group's design, considering aspects like customer experience, the effectiveness of product placements, and the use of AI in retail.

Planner 5D Guide: Setting Up Your Retail Environment Design

[Please note that you may want to use a different AI tool as well.]

1. **Set Up an Account:**
   - Go to Planner 5D and sign up for a free account. (They offer paid features, but the free version has everything you need for this assignment.)
2. **Start a New Project:**
   - Click **Create New Project**, select **Floor Plan**, and adjust dimensions to set up the store space.
3. **Define the Layout:**
   - Use the **Rooms** section to outline the store's shape. Customize wall height and thickness for realism.
4. **Add Fixtures and Product Displays:**
   - Go to **Furniture** and choose categories like **Shelves**, **Counters**, and **Display Units**. Position these elements in key areas, such as around the perimeter or near the checkout area.
5. **Experiment with Product Placement:**
   - Arrange products on shelves and displays based on strategic product assortment and cross-merchandising opportunities (e.g., placing health foods near supplements). Experiment with layouts to optimize flow and customer movement.
6. **Customize the Environment:**
   - Add **Lighting** for ambiance and **Flooring** and **Wall Colours** to reflect the health and wellness theme.
7. **Visualize in 3D:**
   - Click **3D View** to visualize the layout and make adjustments to enhance flow and focus on key product areas.
8. **Share the Project:**
   - Save and share the project via link or download an image for your group presentation.